Zonglin Ye

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Education

University of Electronic Science and Technology of China(UESTC) Chengdu, Sichuan, China
Yingcai Honors College *Sep. 2018 – Jun. 2022*
Bachelor of Engineer, Major in Microelectronic, GPA(3.96/4.0)(90.4 / 100)

University of Electronic Science and Technology of China(UESTC) Chengdu, Sichuan, China
the School of Integrated Circuit Science and Engineering *Sep. 2022 – Present*
Doctor of Philosophy, Major in Electronic Sicence and Technology

Interests

Analog & RF Circuits Design: Frequency Synthesizer, Oscillator, Power Amplifier, Wireless & Wireline Transceiver
Circuits Theory: Dynamical System, Optimization

Award

National Undergraduate Electronics Design Contest 2020 First Prize
China College IC Competition 2023 (IEEE Cup) First Prize (Rank 1)
"Huawei Cup" China Postgraduate IC Innovation Competition 2025 First Prize (Rank 2)
Scholarships From Archiwave Excellent Prize
Suzhou Yucai Scholarship
Scholarships From Silan Microelectronics

Selected Projects

[ISSCC 2026] Low-Spur Fractional-N PLL Based on Harmonic-Mixing Architecture and Spur-Shaping Modulator 2024–2025
- Calibration-free, 46.1 fs jitter, −84.1 dBc integer-boundary spur (1.6 kHz offset)

[CICC 2026] Low-Jitter Multi-Phase PLL Based on High-Mode Ring Oscillator 2024–2025
- 24 fs jitter, 20 GHz, 16-phase outputs

[CICC 2023, JSSC 2024] Time-Amplifier-Based Low-Noise Fractional-N mmWave PLL 2021–2022
- Digital DTC nonlinearity cancellation algorithm
- Dual-DTC precision enhancement technique

[ISSCC 2022, TMTT 2023] Time-Amplifier-Based Low-Noise Integer-N mmWave PLL 2020–2021
- Generalized dynamic analysis and quantitative study of PLL pull-in capability

[CICC 2024, TMTT 2025] Background-Calibrated Fractional-N Octave-Spanning PLL Based on Slide Dithering 2022–2023
- Background DTC gain and nonlinearity calibration using sliding dithering
- High-linearity, low-noise DTC architecture based on self-clamp principle

[TCAS-II 2025] Calibration-Free Octave-tuning Ring-Oscillator-Based Fractional-N PLL Using Harmonic-Mixing Architecture 2023–2024
- Calibration-free, octave-tuning, −80.7 dBc integer-boundary spur (1 kHz offset)

[JOS 2024] Microhertz-Resolution PLL for 4.6 GHz Chip-Scale Atomic Clocks 2020–2021
- Fractional spur cancellation based on digital DLL
- Tunable phase-delay 1PPS synchronizer

Technical Skills

IC Design:
Over 10 tape-out experiences across 180nm, 65nm, 40nm, and 28nm CMOS nodes.
Design experience in PLLs, oscillators, ADCs, PAs, and transceivers.
Proficient in full digital ASIC flow (Cadence); established the lab's first complete frontend and backend flow, including RTL coding, synthesis, UVM verification, formal verification, physical implementation, and signoff timing analysis.
Programming:
Proficient in C/C++, Python, Rust, and Scala; experienced in Linux-based development with GUI frameworks (Qt, Iced).
Proficient in electrostatic simulation; designed and implemented a BEM-based RC extraction tool using Rust+Python, achieving $< 1\%$ error compared to EMX.
Over 6 years of experience in deploying and maintaining Slurm-based HPC clusters.
Embedded Systems:
Experienced in embedded Linux development and embedded design under ARM/RISC-V/Xtensa Socs.
Developed automated test frameworks and chip characterization programs with VISA-compliant instruments.
Designed Ethernet-to-SPI debug bridges and multi-channel chip monitoring/testing suites.
Others:
Extensive PCB design experience with 100+ boards completed.
Proficient in thermal simulation platforms; performed CFD and heat transfer analysis for computational unit air-cooled thermal design.

Selected Prublications

[1] **Z. Ye**, Y. Sun, L. Hou, Y. Ding, Y. Wen, H. Zhang, X. Geng, Q. Xie, S. Yang, Z. Wang , "A 20-GHz Frequency Synthesizer with Spur-Shaping Modulator Achieving 46.2-fs Jitter and –76.5 dBc Worst-Case Fractional Spur", *ISSCC2026*
[2] **Z. Ye**, L. Hou, Y. Sun, Y. Ding, Y. Wen, H. Zhang, H. Luo, Y. Yang, C. Zhang, X. Geng, Q. Xie, C. Wang, J. Zhou, Z. Wang , "A 28nm 11.2-to-20-GHz PLL-based 16-Phase Clock Generator with High-Order-Mode Ring Oscillator Achieving 24.3-fs Jitter for 320-Gsps-Sample-Rate Data Converter", *CICC2026*
[3] **Z. Ye**, X. Geng, Z. Shi, H. Zhang, L.Hou, Q. Xie and Z. Wang, "A 6.8-to-14.4-GHz Fractional-N CPPLL Featuring a Slide-Dithering-Based DTC Nonlinearity Calibration With Self-Clamping DTC Achieving -65.8-dBc Fractional Spur at 1.6-kHz Offset", *TMTT2025*
[4] **Z. Ye**, H. Zhang, Y. Sun, X. Geng, Q. Xie, S.Yang and Z. Wang, "A 6.8-14.4GHz Calibration-Free Fractional-N RO-Based Harmonic-Mixing Frequency Synthesizer Achieving -80.7dBc Fractional Spur", *TCASII2025*
[5] **Z. Ye***, X. Geng*, Y. Xiao, Q. Xie and Z. Wang, "A Sub-50-fsrms Jitter Fractional-N CPPLL Based on a Dual-DTC-Assisted Time-Amplifying Phase-Frequency Detector With Cascadable DTC Nonlinearity Compensation Algorithm", *JSSC2024*
[6] **Z. Ye***, X. Geng*, Z. Shi, H. Zhang, Q. Xie and Z. Wang, "A 6.8-to-14.4GHz Octave-Tuning Fractional-N Charge-Pump PLL with Slide-Dithering-Based Background DTC Nonlinearity Calibration for Near-Integer Fractional Spur Mitigation Achieving 78fs RMS Jitter and -258.6dB FoMT", *CICC2024*
[7] X. Geng*, **Z. Ye***, Y. Xiao, Q. Xie and Z. Wang, "A 26GHz Fractional-N Charge-Pump PLL Based on A Dual-DTC-Assisted Time-Amplifying-Phase-Frequency Detector Achieving 37.1fs and 45.6fs rms Jitter for Integer-N and Fractional-N Channels", *CICC2023*
[8] X. Geng, **Z. Ye**, Y. Xiao, Y. Tian, Q. Xie, Z. Wang, "A 25.8-GHz integer-N CPPLL achieving 60-fs rms jitter and robust lock acquisition based on a time–amplifying phase–frequency detector", *TMTT2023*
[9] X. Geng, Y. Tian, Y. Xiao, **Z. Ye**, Q. Xie, Z. Wang, "A 25.8GHz Integer-N PLL With Time-Amplifying Phase-Frequency Detector Achieving 60fsrms Jitter, -252.8dB FoMJ, and Robust Lock Acquisition Performance", *ISSCC2022*
[10] H. Heng, L. Jia, **Z. Ye**, K. Wang, Q. Xie, J. Zhou, Z. Wang, "A Broadband Digital Power Amplifier With 1-dB Psat Bandwidth of 5.5–9.5 GHz", *MWCL2024*
[11] H. Zhang, X. Geng, **Z. Ye**, K. Wang, Q. Xie, Z. Wang, "A frequency servo SoC with output power stabilization loop technology for miniaturized atomic clocks", *JOS2024*
[12] Z. Zhang, Z. Zhao, **Z. Ye**, K. Wang, Q. Xie, Z. Wang, "A compact lumped-element two-way differential to single-ended Wilkinson power combiner with embedded impedance transformation", *MOTL2023*